

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. What distinguishes a congenital (true) diverticulum from an acquired one, structurally?
2. Where in the GIT is the commonest acquired diverticulum found, and why does it form there?
3. List three of the “rule of 2’s” facts about Meckel diverticulum.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

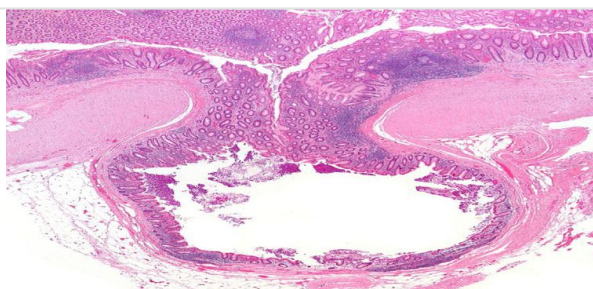
1 Diverticulum — the frame

What	A blind outpouching of the mucosa in, or through, the muscularis propria.
Congenital (true) diverticulum	Lined by all three (four) layers of the gut wall (mucosa, submucosa, muscularis propria, serosa).
Acquired (false) diverticulum	Either lacks muscularis propria entirely , or has an attenuated muscularis propria.
Where	May develop in any part of the GIT.
Commonest true congenital diverticulum	Meckel diverticulum.
Commonest site of acquired diverticula	Sigmoid colon.

2 Acquired diverticulum (sigmoid colon)

Mechanism	Develops from increased intraluminal pressure forcing mucosa/submucosa through weak points in the wall; increased pressure is linked to a low-fiber diet .
Weak points	Where the inner circular muscularis propria is penetrated by nerves, arterial vasa recta , and their connective-tissue sheaths.
Clinical features	Often asymptomatic · abdominal pain · constipation and/or diarrhea.

Complication	Detail
Diverticulitis	Inflammation & abscess formation
Intestinal obstruction	By scarring
Fistula formation	
Perforation and peritonitis	
Bleeding	



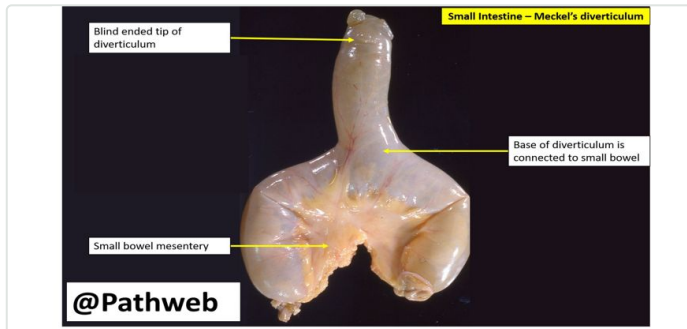
Sigmoid colon diverticulum: mucosa/submucosa invaginating through the muscularis propria

3 Meckel diverticulum — illness script

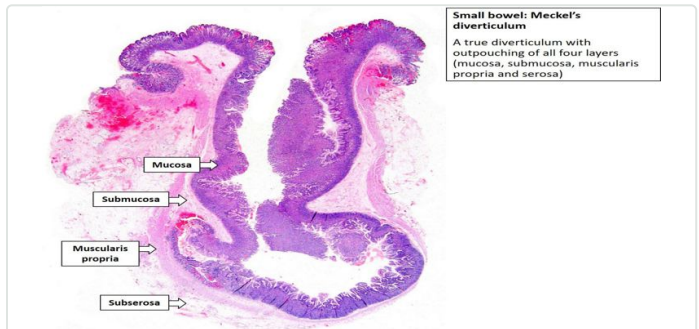
Also known as	True congenital diverticulum of the small bowel.
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WHY it happens	Failure of involution of the vitelline duct , which connects the lumen of the developing gut to the yolk sac.
WHERE	Extends from the antimesenteric border of the ileum.
WHAT you see	Mucosal lining may resemble normal small intestine, or contain ectopic pancreatic or gastric tissue , which may cause inflammation, ulceration, bleeding, and obstruction.

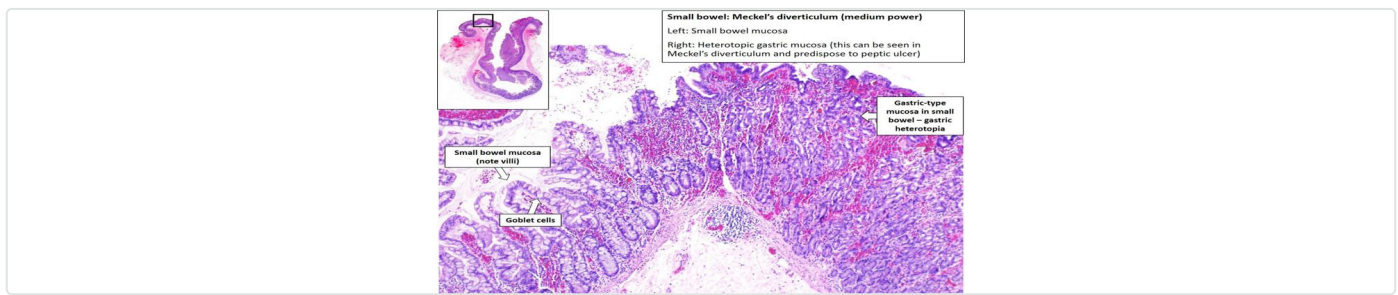
The rule of 2's: occurs in ~2% of the population · located within **2 feet (60 cm)** of the ileocecal valve · ~**2 inches (5 cm)** long · **twice as common in males** · most often symptomatic by age **2 years** · **2/3** have ectopic tissue (gastric or pancreatic) · only **2%** become symptomatic.



Meckel diverticulum: blind-ended tip, connected to the small bowel at its base



Meckel diverticulum histology: a true diverticulum with all four wall layers



Meckel diverticulum: small bowel mucosa (left) alongside heterotopic gastric mucosa (right), which predisposes to peptic ulceration

P Must-remember pearls

- **Congenital diverticula are true** (all wall layers); **acquired diverticula lack or have attenuated muscularis propria**.
- **Meckel = commonest true congenital diverticulum; sigmoid colon = commonest site of acquired diverticula**.
- Acquired (sigmoid) diverticula form at weak points where the circular muscularis propria is penetrated by **vasa recta and nerves**, favored by a **low-fiber diet**.
- Meckel diverticulum comes from failed **vitelline duct** involution, arising from the **antimesenteric** border; ectopic **gastric mucosa** in it predisposes to peptic ulceration.
- The **rule of 2's** for Meckel: 2% prevalence · within 2 feet of the ileocecal valve · 2 inches long · 2× more common in males · symptomatic by age 2 · 2/3 have ectopic tissue · 2% become symptomatic.

Q Test yourself

1. A colonic diverticulum is found to lack a muscularis propria layer entirely. This is best classified as:

- A. An acquired (false) diverticulum
- B. A congenital (true) diverticulum
- C. A Meckel diverticulum
- D. A duplication cyst
- E. A pseudopolyp

2. Sigmoid diverticula characteristically form at the point where the circular muscularis propria is penetrated by:

- A. The inferior mesenteric vein
- B. Peyer patches
- C. The appendices epiploicae
- D. Vasa recta and their accompanying nerves
- E. Taenia coli

3. Meckel diverticulum results from failure of involution of which embryologic structure?

- A.** Urachus
- B.** Processus vaginalis
- C.** Vitelline duct
- D.** Notochord
- E.** Cloacal membrane

4. A 2-year-old presents with painless rectal bleeding. A Meckel diverticulum is resected and found to contain ectopic tissue, which predisposed to ulceration and bleeding. This ectopic tissue is most likely:

- A.** Squamous esophageal mucosa
- B.** Hepatocytes
- C.** Renal tubular tissue
- D.** Thyroid follicles
- E.** Gastric mucosa

5. Which of the following is a recognized complication of sigmoid (acquired) diverticular disease?

- A.** Aganglioneurosis
- B.** Diverticulitis with abscess formation
- C.** Carcinoid syndrome
- D.** Napkin-ring constriction
- E.** Gastric heterotopia

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 • A — Acquired diverticula either lack muscularis propria entirely or have an attenuated layer, unlike true congenital diverticula which retain all wall layers.

2 • D — The weak points of the colonic wall are where the inner circular muscularis propria is penetrated by nerves, arterial vasa recta, and their connective tissue sheaths.

3 • C — Meckel diverticulum occurs due to failure of involution of the vitelline duct, which connected the developing gut lumen to the yolk sac.

4 • E — About two-thirds of Meckel diverticula contain ectopic gastric or pancreatic tissue; ectopic gastric mucosa can cause acid-related ulceration and bleeding.

5 • B — Complications of acquired diverticular disease include diverticulitis and abscess, obstruction from scarring, fistula, perforation/peritonitis, and bleeding.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pathology GIT Block · MEDucation by Mattin Majid · summarised from SGL3, Dr Jawher Tahir Omer, Hawler Medical University
Study alongside the lecture, not instead of it.